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IS4301 Agile IT with DevOps – Lecture 1

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Department of Information Systems and Analytics
National University of Singapore

Learning Objectives

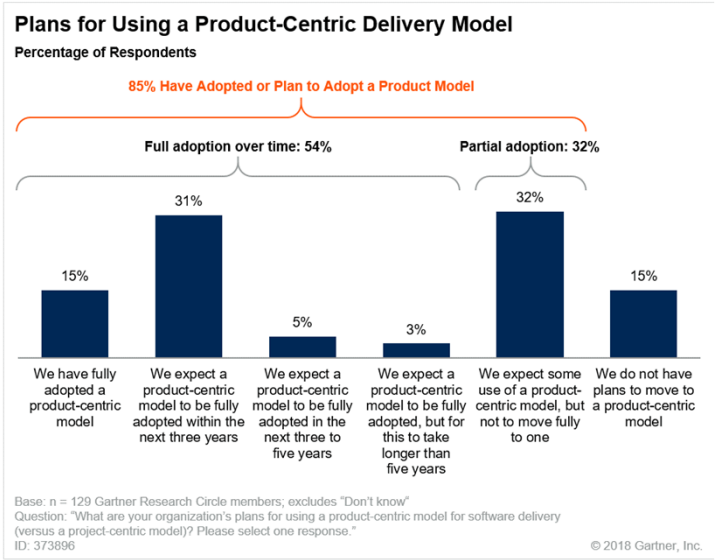
At the end of this lecture, you will understand:

- Introduction to course syllabus.
- Why Agile?



Gartner Survey Finds 85 Percent of Organizations Favor a Product-Centric Application Delivery Model, 2019.

- Eighty-five percent of organizations have adopted, or plan to adopt, a product-centric application delivery model, according to a survey* by Gartner, Inc. Although full adoption is rare, overall, survey respondents use the product-centric model for 40 percent of their work in 2018. Gartner predicts that this figure will reach 80 percent by 2022.
- "The increase in how quickly and broadly organizations are adopting the product-centric application model doesn't arise randomly. It goes hand-in-hand with the adoption of agile development methodologies and DevOps," said Bill Swanton, distinguished research vice president at Gartner. "In addition, an increasing number of applications that IT teams develop are used by external parties, such as clients or partners, and require the increased customer focus that characterizes the product-centric model."
- Name me an organization who has not adopted agile in this age.



Course Timings

- Course: ~~Wed 9-11 am~~ Wed 10-1130 am
- Location: LT15

- Tutorial location :
  - Tutorial Group 1: ~~Wed 1100-1200 hrs,~~ Wed 1130-1230, LT15
  - ~~Tutorial Group 2: Wed 1200-1300 hrs, LT15~~



What Will You Learn in this Module?

Topics:

- Traditional SDLC software development and operations vs. Agile
- Overview of Agile Methods, team structure, lifecycle, technical practices, user stories, agile estimation
- Case studies in banking and other industries
- Overview of DevOps, Continuous Integration Continuous Deployment (CI/CD)
- Overview of foundational technologies, platform and open toolchains
- Developing simple “Hello World” microservices and DevOps – This is not a DevOps Toolchain dev course

What Do You Read in this Module?

- Textbook for this module:
  - No textbook
- Real World Case Studies especially banking industry
- Instructor-led, e-Learning and peer learning
- Learn and practice approach
- Wikis, githubs, ...

Syllabus

Lecture 1: Course Introduction to Agile IT with DevOps
Intro to course syllabus. Digital transformation and Agile.

Lecture 2: Overview of Agile Methods and DevOps
Traditional SDLC software development and operations vs. Agile Methods, impact to business and IT agility, Agile Manifesto, Agile Development Introduction, Agile Workspace, Case Studies including Banking.

Lecture 3: Understand essential concepts of different Agile Methods and Roles in Agile Team Structure.

Tutorial 1: Formation of tutorial teams & selection of Fintech Challenge Statement

Lecture 4: Understand Agile lifecycle, ceremonies and technical practices.

Tutorial 2: Understand roles in sprint teams, formation of sprint team roles, conduct stand-up and sprint ceremonies

Syllabus

Lecture 5: Understand User stories, agile estimation. **Tutorial 3: Estimation and develop user stories**
Lecture 6: Understand user story acceptance criteria and testing, sprint planning, backlog grooming **Tutorial 4: Sprint planning**
Recess and Exam Week – No lecture
Lecture 7: **Monolithic vs. Microservice Architecture, Overview of Microservice architecture** **Tutorial 5: Lab exercise: Simple "Hello, World" application with agile, User authentication, intro to client-side, server-side, version control**
Lecture 8: Understand Microservice architecture (characteristics, languages, architectural style, ReST APIs, API gateway, service registry and discovery, monitoring) **Tutorial 6: Lab exercise: getting started with microservices for server-side implementation**

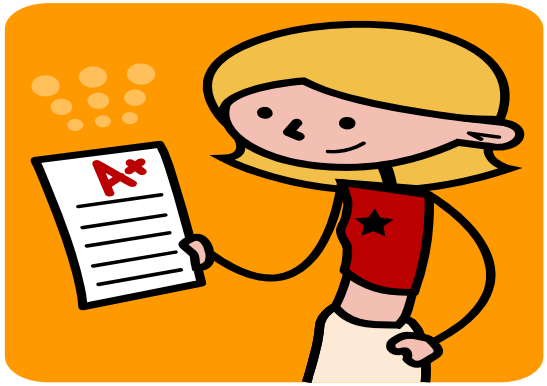
Syllabus

Lecture 9: Introduction of DevOps, Overview of CI/CD, Continuous Delivery, Continuous Deployment, Continuous Integration, Value, toolchain **Lab exercise: CI/CD with Git Actions**
Lecture 10: Containerization, Virtualization, Standards **Tutorial 8: Lab exercise: Containerization with Docker**
Lecture 11: Agile and DevOps case studies (AWS, Netflix, Airbnb, DBS) **Agile Adoption at scale case studies in banks.** **Tutorial 9: Prepare for Project Presentation**
Lecture 12: Project Presentations Week during lecture and tutorial sessions **Submission of Individual Written Assignment.**
Lecture 13: Industry experience case study and Career tips with Q&A

Might have quiz last session

How Will You be Tested?

- Grading Scheme:
  - Evaluation of individual presentation in tutorials, participation, final graded quiz (20%)
  - Individual written assignment (30%)
  - Final Project Team Proposal Presentation (50%)
  - Evaluation of contributions by peers in same team using TeamMates (0 %)
- Tutorial Sessions
  - Incremental project preparations of topic covered in weekly lessons
  - Research after each weekly lessons
- Final project presentations on Nov 5th during class and tutorial sessions
  - 15 mins for each project team
- Submission of individual written assignment on Nov 5th.
- Final graded quiz on Nov 12th 5% of first bullet point



Grading of Project Presentation

- Introduction – 5%
- Demo – 25%
- Innovation – 25%
- Conclusion – 5%
- Agile practices – 25%
- Special efforts – 10% Coding and research
- Organization – 5%

Common Questions

- Can I choose my project team members? yes
- What is the size of a project team? 4-5
- What will make an A+ in this course? be interested and do own research
- Does contents of presentation matter more than presentation skills? Both
- Can I choose topics for team project? Yes
- How does peer evaluation work?
- Is it possible that team members will receive different grades within the same team?
- Are attendance and participation important? yes
- Will lectures be recorded for replay if I am unable to attend classes? Yes but still need to come for class
- What is the individual written assignment? Open book, 1 - 1.5mth duration
- What is the level of programming in this course? low level programming
- What time does class starts?

9/9.10

What are the Rules?

- If you don't understand something, please ask. Participation is an important aspect of learning.
- If you send me an email (disfsb@nus.edu.sg), start subject title with "IS4301:". For pirority queue



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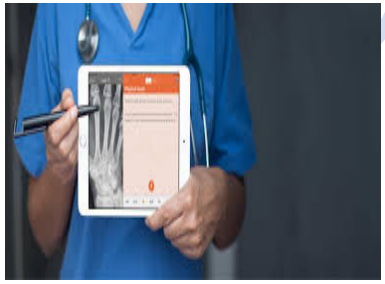
About Myself - Always learning …

Formative 1990-1999

- Teaching and Research (NUS ISS, UT Austin)
- Fighter Jet Simulator
- Banking IT Ops and Innovation
- Entrepreneur Dr. World/Raffles Medical – Mobile EMR and CMS







From 2000

Advisory Boards:

- Keppel Corp Technology Advisory Panel
- National Research Foundation Advisory Comm, Central Gap Fund
- NUHS Group IT Committee
- Singapore Poly EEE Dept Advisory Comm
- Chairman NICF Steering Comm
- ISCA IT Adv Comm Chairman, CFO Comm

- IBM – first Software Architect India/South Asia, first CTO ASEAN, first IBM Distinguished Engineer in AP
- Adjunct Assoc Prof, Adunct Professor
- Global Head T&O Digital Transformation (Retail Banking, Private Banking and Wealth Management)
- Deputy Chief Executive (Services, Governance, CyberSecurity)
- Managing Director and Group Head, Technology Services, T&O
- NUS National University of Singapore Professor (Practice)

IBM | NUS National University of Singapore | Standard Chartered | GOVTECH SINGAPORE | DBS Live more, Bank less

About TA

Ding Jingqi

- Second-year PhD student in Information Systems
- Email: jingqi.ding@u.nus.edu
- Lab: COM3-02-65 - ISA Lab 1

Research Interests

- Digital resilience, digital transformation, computational social science

Personal Interests

- Play with my dogs, play ukulele, play tennis



Learning Objectives

At the end of this lecture, you will understand:

- Introduction to course syllabus.
- Why Agile?



Example: A 4-5 month internship

- Scope
  - Aggregate business performance datasets from 50+ spreadsheets and 5 databases and display onto dashboards for decision making.
- Resources
  - Two interns – you and another intern from University X.
  - Industry supervisor is busy most of the time but is committed to help you with questions.
- How would you plan and execute for an A+ outcome from this internship?

Need to learn to streamline the user stories. So start with user experience, how does the user interact with the dashboards. Start with user experience. So figure out what the user wants to see which will determine the data you will get.

Example: First Job as a Fresh Grad in a Large Enterprise

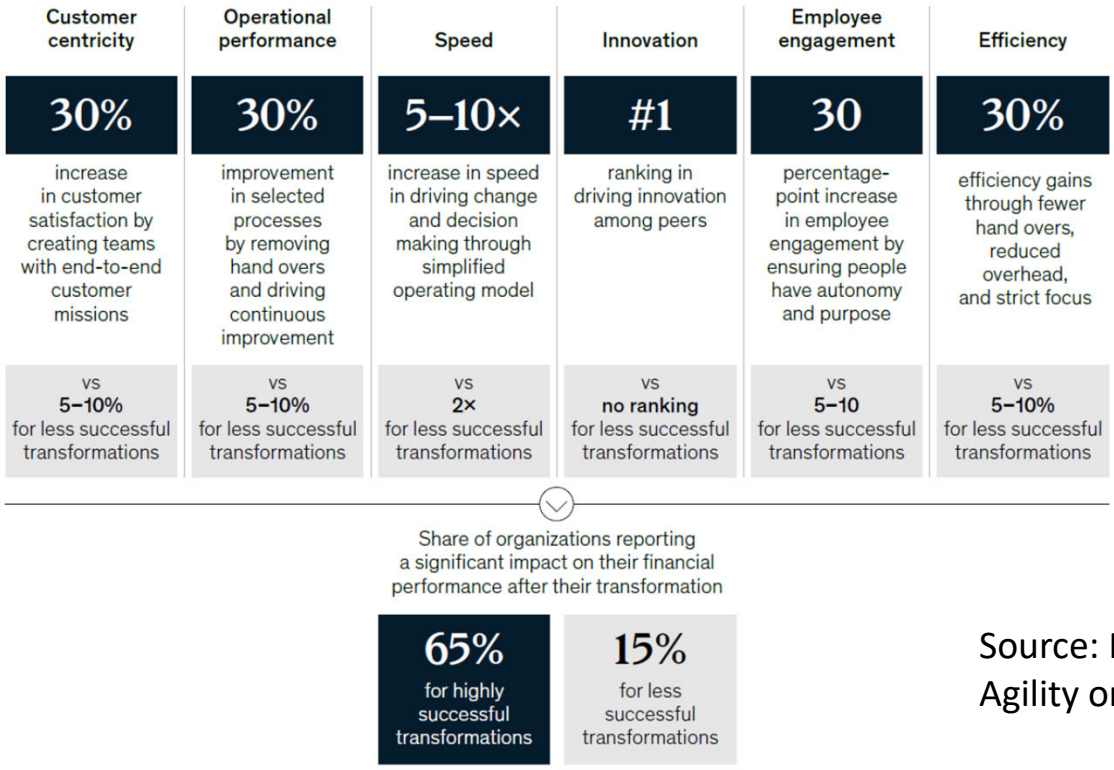
- Scope
  - Member of a Digital Banking app of a large Bank to deliver top notch digital banking services within a legacy systems-heavy IT environment in fast-paced settings.
  - Migrate banking services from legacy to microservices
  - Integrate digital services with many other backend services
  - Data and AI driven services for digital banking app
  - AI-assisted development
- Resources
  - Your team members come with different strengths, experiences.
  - Not enough resources. Everyone works super hard.
  - Your manager and business stakeholders called themselves agile squads but you and your team are asked to deliver all services within fixed time and budget.
  - Many extended teams to work with, e.g., mainframe team, infrastructure-as-a-service team, payment team, customer journey experience team, risk and audit team, enterprise architecture team, data and AI team, and so on.
  - Changing and evolving job roles due to AI and global landscape
- What skills are needed for you to be successful as a fresh grad?

Example: Job in a Small-Medium-Enterprise (SME)

- Scope
  - Develop product roadmap and implementation for fast go-to-market before competition
  - Pitch to investors
  - Manage cash prudently
- Resources
  - Your team members come with different strengths, experiences.
  - Not enough resources. Everyone works super hard.
  - Business development, developer, architect, operations all rolled into one
- What skills are needed for you to be successful as a fresh grad?

Highly successful agile transformations result in a step change in performance, with impact achieved in multiple dimensions.

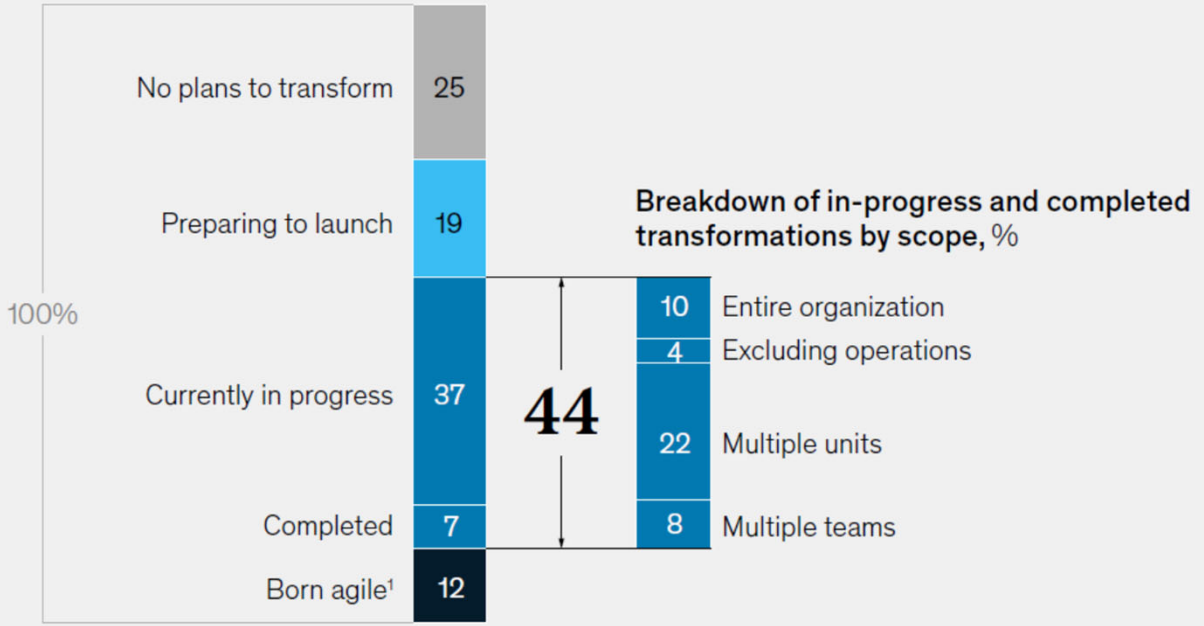
Impact achieved by highly successful[1] agile transformations



Source: McKinsey on Impact of Agility on organizations

Forty-four percent of respondents report agile transformations at their organizations, but only 14 percent are enterprise-wide.

State of agile transformation, % of respondents (n = 2,190)



[1]Organizations in which agile practices and concepts have always been core to how they work.

Source: McKinsey on Impact of Agility on organizations

What's in Next Lecture?

- Overview of Agile Methods and DevOps
  - Traditional SDLC software development and operations vs. Agile Methods,
  - impact to business and IT agility,
  - Agile Manifesto,
  - Agile Development Introduction,
  - Agile Workspace,
  - Case Studies including Banking.